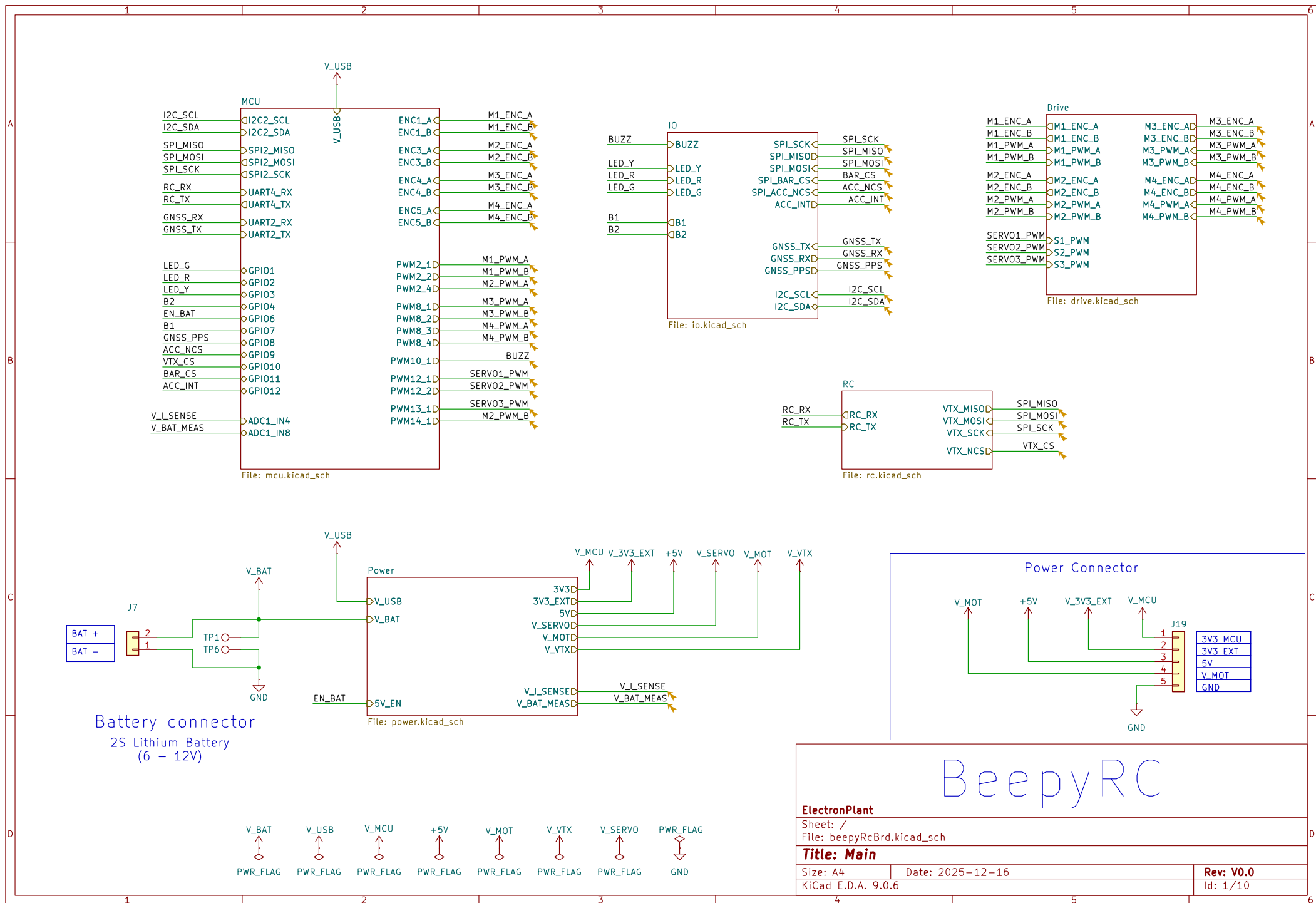
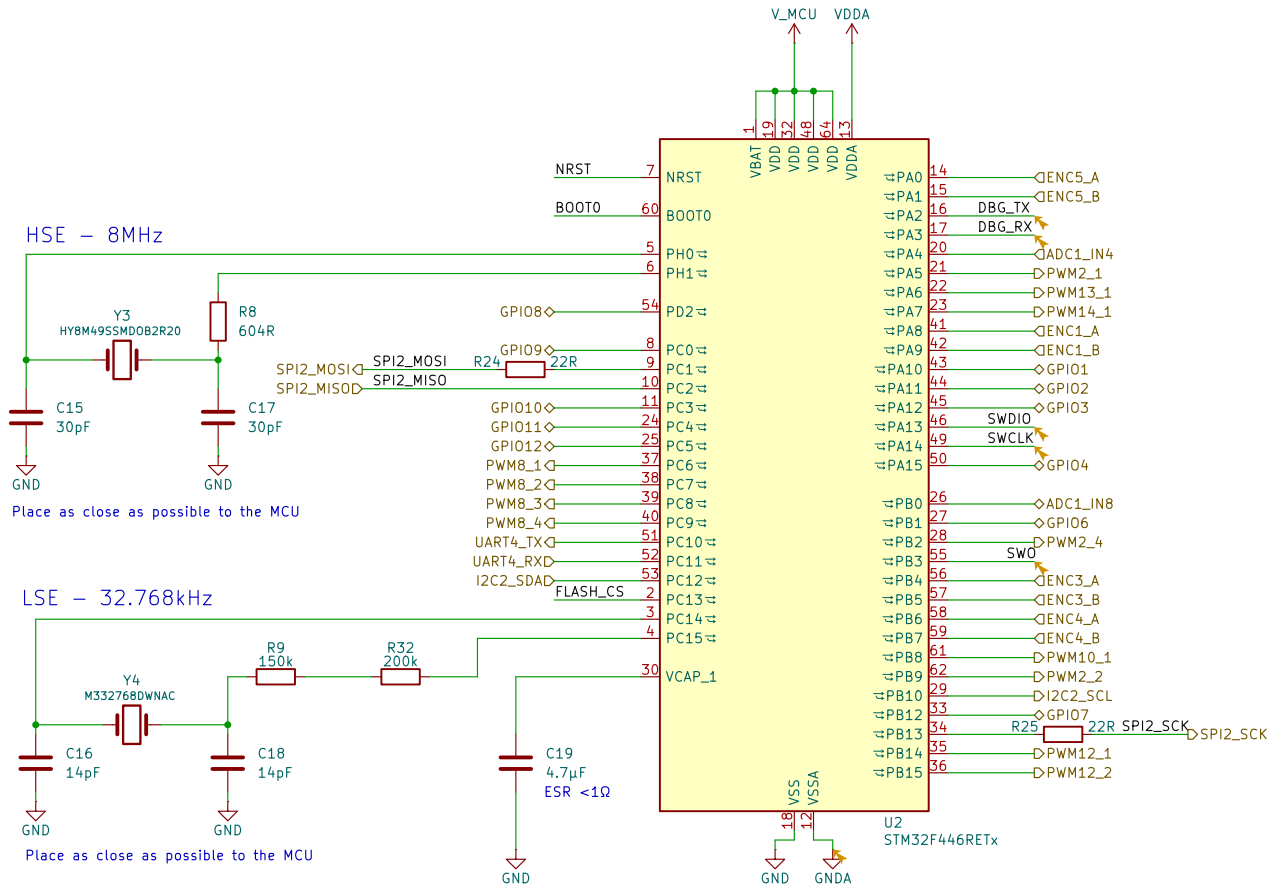
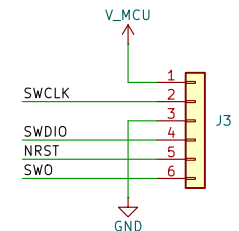






- 10uF ceramic cap for the package.
- 100nF ceramic between pins 18-19, 31-32, 47-48 and 63-64.
- Vbat on pin 1 is connected to VDD with a 100nF decoupling cap.
- VDDA is connected to VDD through a ferrite bead.
- VDDA is decoupled with a 1uF and a 100nF capacitor between pins 12-13.
- GND is directly connected to the GND ground plane.

Pin 13 is connected to the UART2_TXD pin (R22) and the V_USBD pin. The diagram shows a green wire connecting the TXD pin to the V_USBD pin, which is then connected to GND.

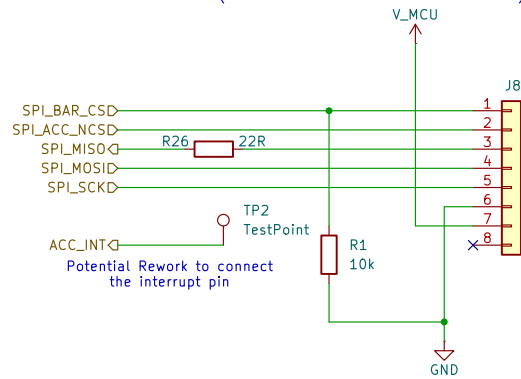


BeepyRC

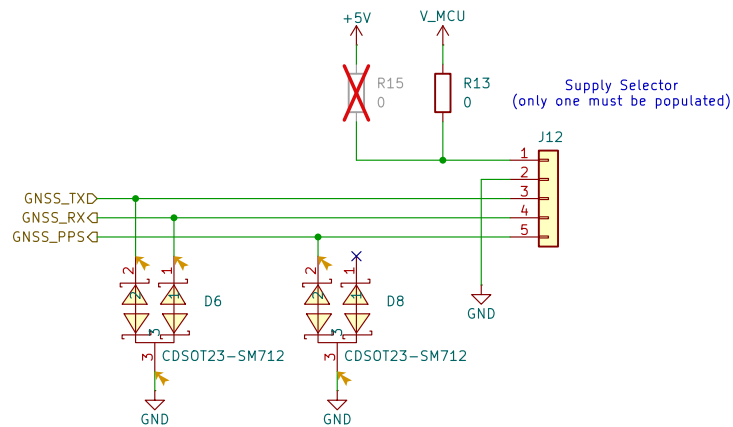
Id: 2/10

Sensors

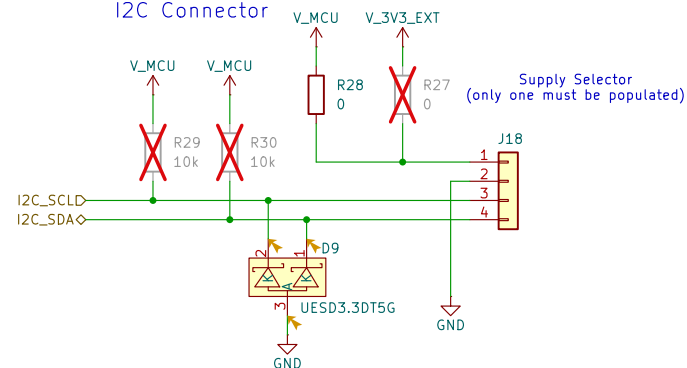
GY-91 (9-axis MPU9250 + BMP280) Module



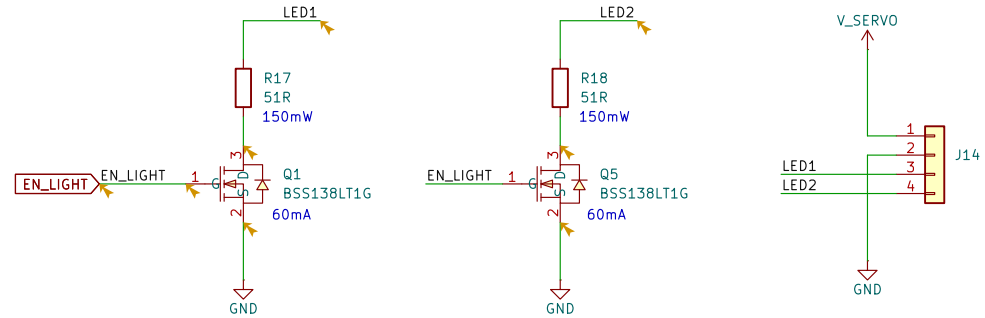
GNSS Module (L76K/VG7669T160N0MA)



I2C Connector

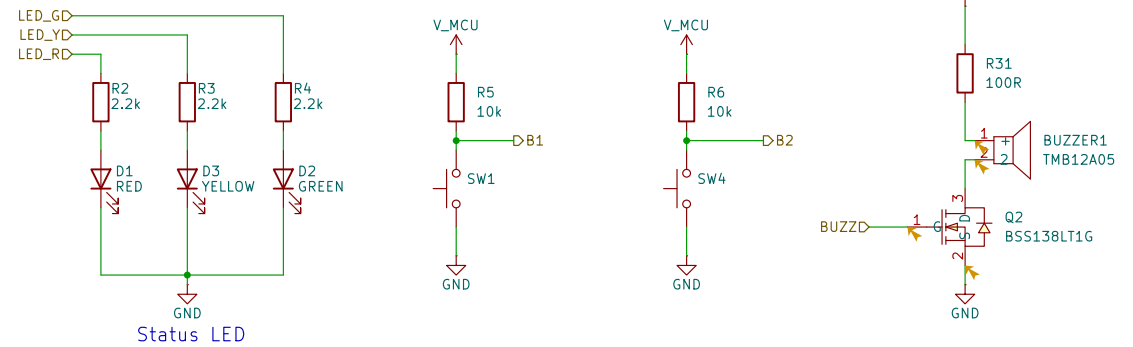


Lights



Current limiting resistor sizing:
 - LED forward voltage: ~3V (2.8 - 3.4)
 - Max forward current: 60mA

User Interface



BeepyRC

ElectronPlant

Sheet: /IO/

File: io.kicad_sch

Title: IO

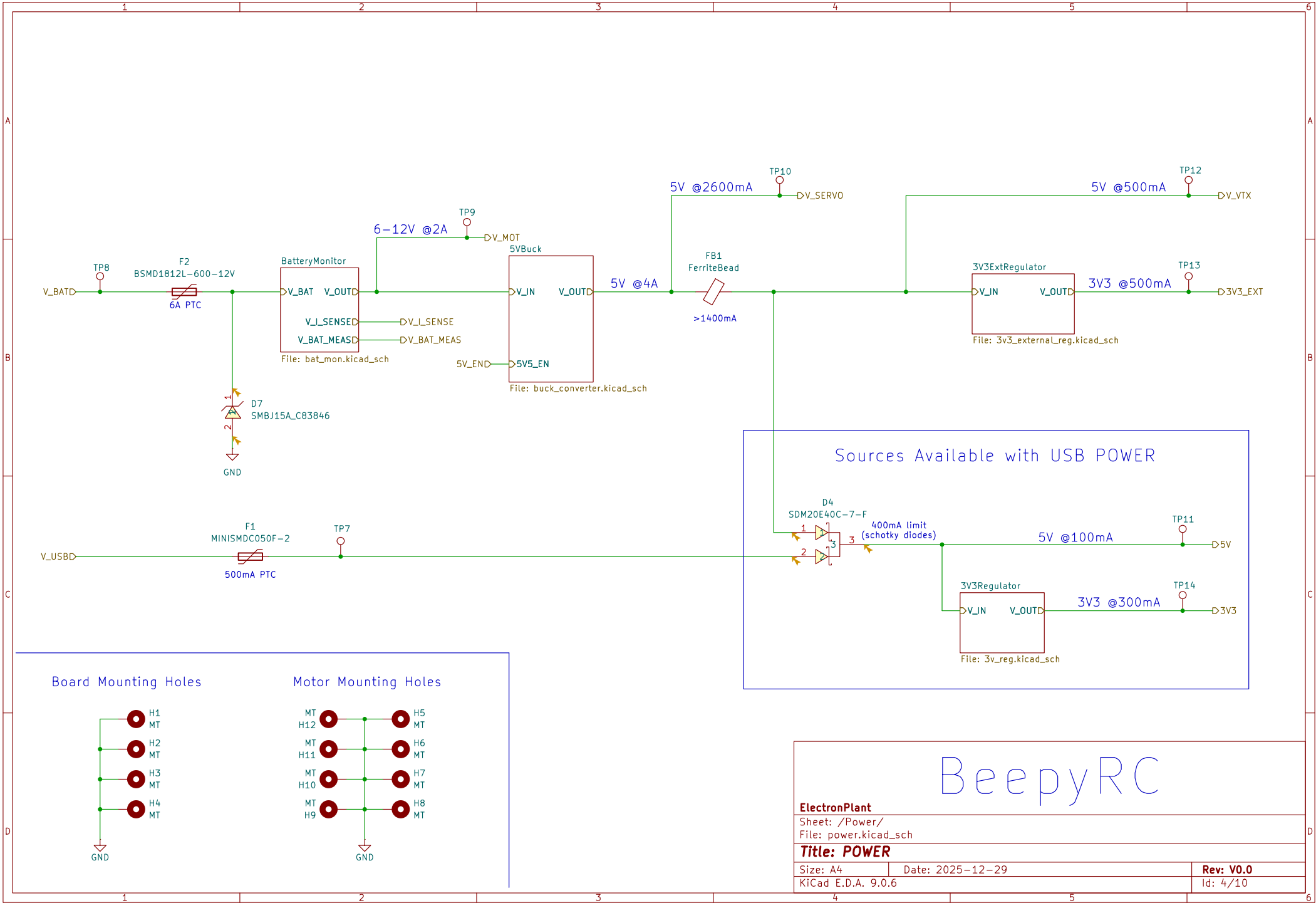
Size: A4

Date: 2025-12-29

Rev: V0.0

KiCad E.D.A. 9.0.6

Id: 3/10



BeepyRC

ElectronPlant

Sheet: /Power/
File: power.kicad_sch

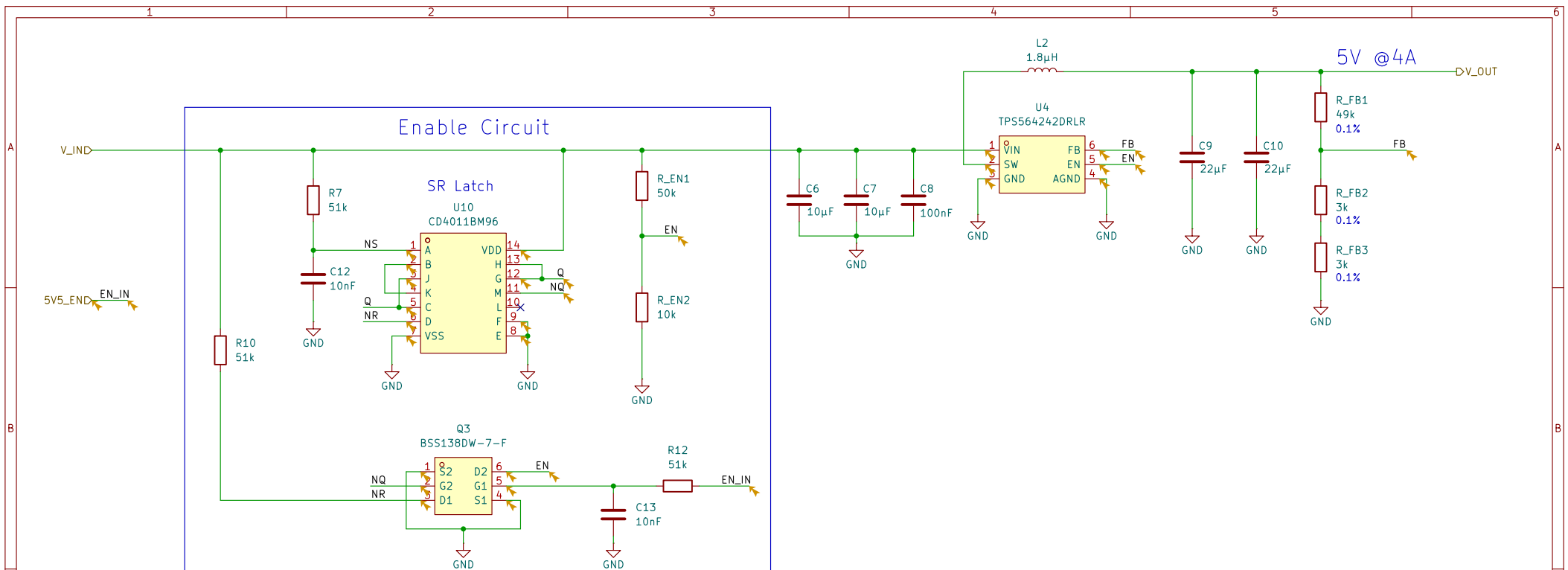
Title: POWER

Size: A4 Date: 2025-12-29

KiCad E.D.A. 9.0.6

Rev: V0.0

Id: 4/10



Notes

--- Output capacitance ---

$$C_{OUT} = 44\mu F \text{ (} 22\mu F, 100\mu F \text{) [typ., (min, max)]}$$

$$I_{L_CAP} = (V_o * (V_{in} - V_o)) / (\sqrt{12} * V_{in} * L * f_{sw}) = 0.39A$$

L -> Inductor value
fsw -> Switching frequency (1200kHz)

--- Inductance ---

$$L (V_o = 5V) = 1.8 \text{ (} 1.5, 4.7 \text{) } \mu H \text{ [typ., (min, max)]}$$

Inductance Ratings:

$$I_{p-p} = (V_o / V_{in_max}) * (V_{in_max} - V_o) / (L * f_{sw}) = 1.35A$$

$$I_{peak} = I_o + I_{p-p} / 2 = 5.35A$$

$$I_{L_RMS} = \sqrt{I_o^2 + (1/12) I_{p-p}^2} = 4.3A$$

--- Feedback Resistors ---

$$V_o = 0.6 * (1 + R_{FB1} / R_{FB2})$$

$$R_{FB2} = (0.6 + 0.6 R_{FB1}) / V_o$$

--- EN input ---

When:

- EN is > 1.18(1.07, 1.33)V the module is enabled.
- EN is < 1(0.95, 1.2)V the module is disabled.

$$\text{Fixing } R_{EN1} + R_{EN2} \rightarrow R_{EN2} = (R_{EN1} + R_{EN2}) * V_{EN} / V_{IN}$$

$$V_{CUT} = V_{EN} * (R_{EN1} + R_{EN2}) / R_{EN2}$$

To limit to 6V input with $R_{EN1} + R_{EN2} = 50k$:

$$R_{EN2} \sim 8.2k$$

$$R_{EN1} \sim 40k$$

$$V_{CUT} = 5.87V \text{ (} 5.58, 7.05 \text{)}$$

This is too unreliable so it will be driven by the MCU!!

BeepyRC

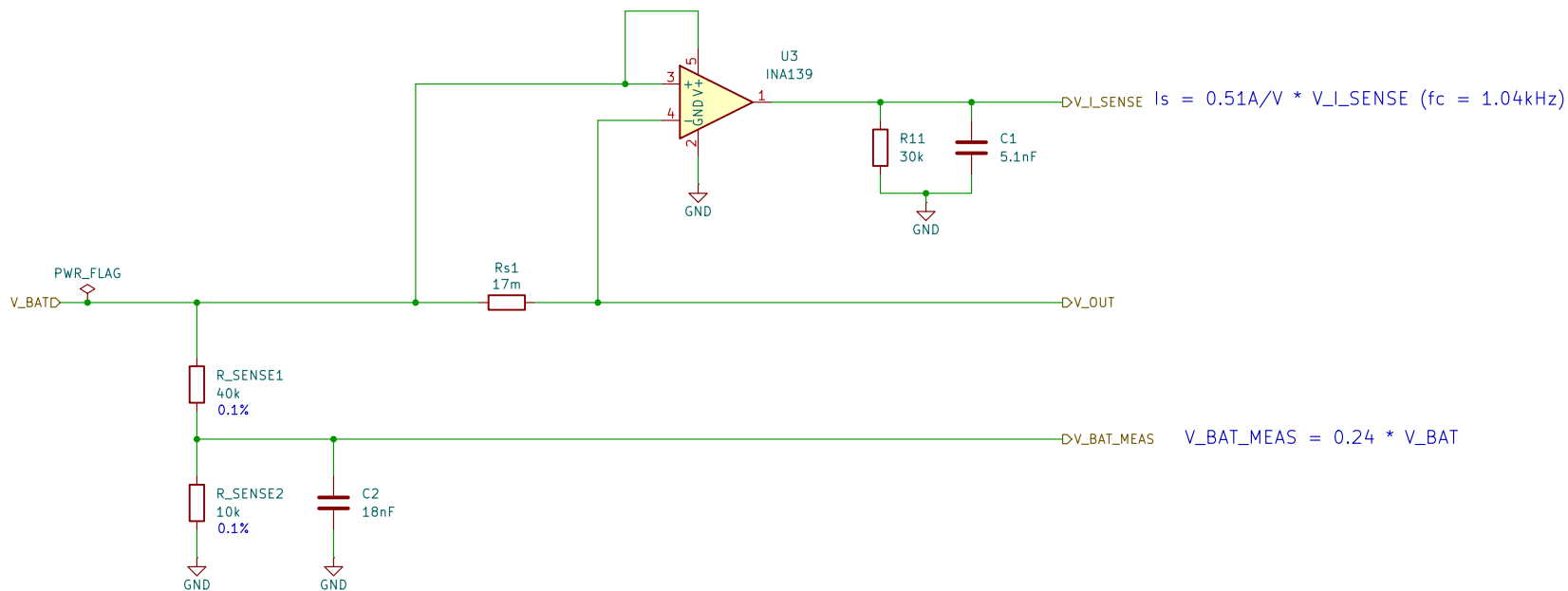
Electron Plant

Sheet: /Power/5VBuck/
File: buck_converter.kicad_sch

Title: Buck Converter

Size: A4 Date: 2026-02-08
KiCad E.D.A. 9.0.6

Rev: V0.0
Id: 5/10



--- Battery Voltage ---

2S battery range 6 – 8.4V, with 7.4 nominal
 Maximum ranges 6 – 12V --> Limited by the N20 motors
 R_SENSE is set to:

1. Be <50kOhm (ADC external resistance limit),
2. V_BAT_MEAS is ~3V @ 12V
3. Low-pass filter with Fc ~1kHz

--- INA139 ---

Output Voltage: $V_o = I_s * 1mA/V * R_L$
 V_s full scale = 100mV (could also be 50mV to 500mV)
 R_L -> Set to generate ~3V at 6A (<50kOhm, ADC external resistance limit)
 C_L -> Low pass filter with a cut-off frequency at ~1kHz

--- Shunt Resistor ---

$V_s = 100mV$ with max current
 $\max(I_s) = 6A > (4 * 0.35A \text{ for the motors} + 4A \text{ for the } 5.5V \text{ rail})$
 $R_s = V_s / \max(I_s) = \sim 17mOhm$
 Power Rating: 0.6W

BeepyRC

Electron Plant

Sheet: /Power/BatteryMonitor/
 File: bat_mon.kicad_sch

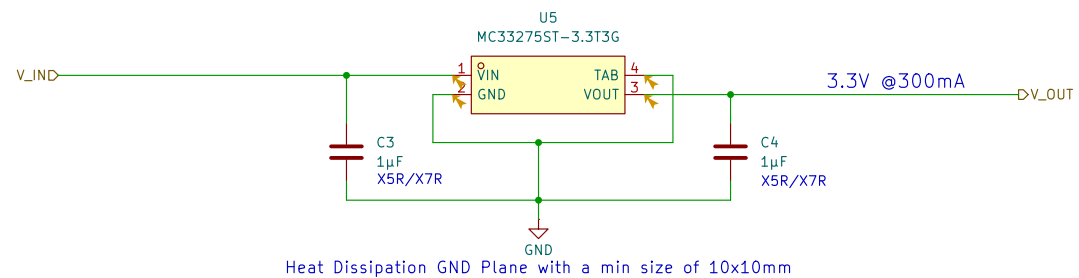
Title: Battery Monitor

Size: A4 Date: 2026-02-11

KiCad E.D.A. 9.0.6

Rev: V0.0

Id: 6/10



BeepyRC

ElectronPlant

Sheet: /Power/3V3Regulator/
File: 3v_reg.kicad_sch

Title: 3V3 Regulator

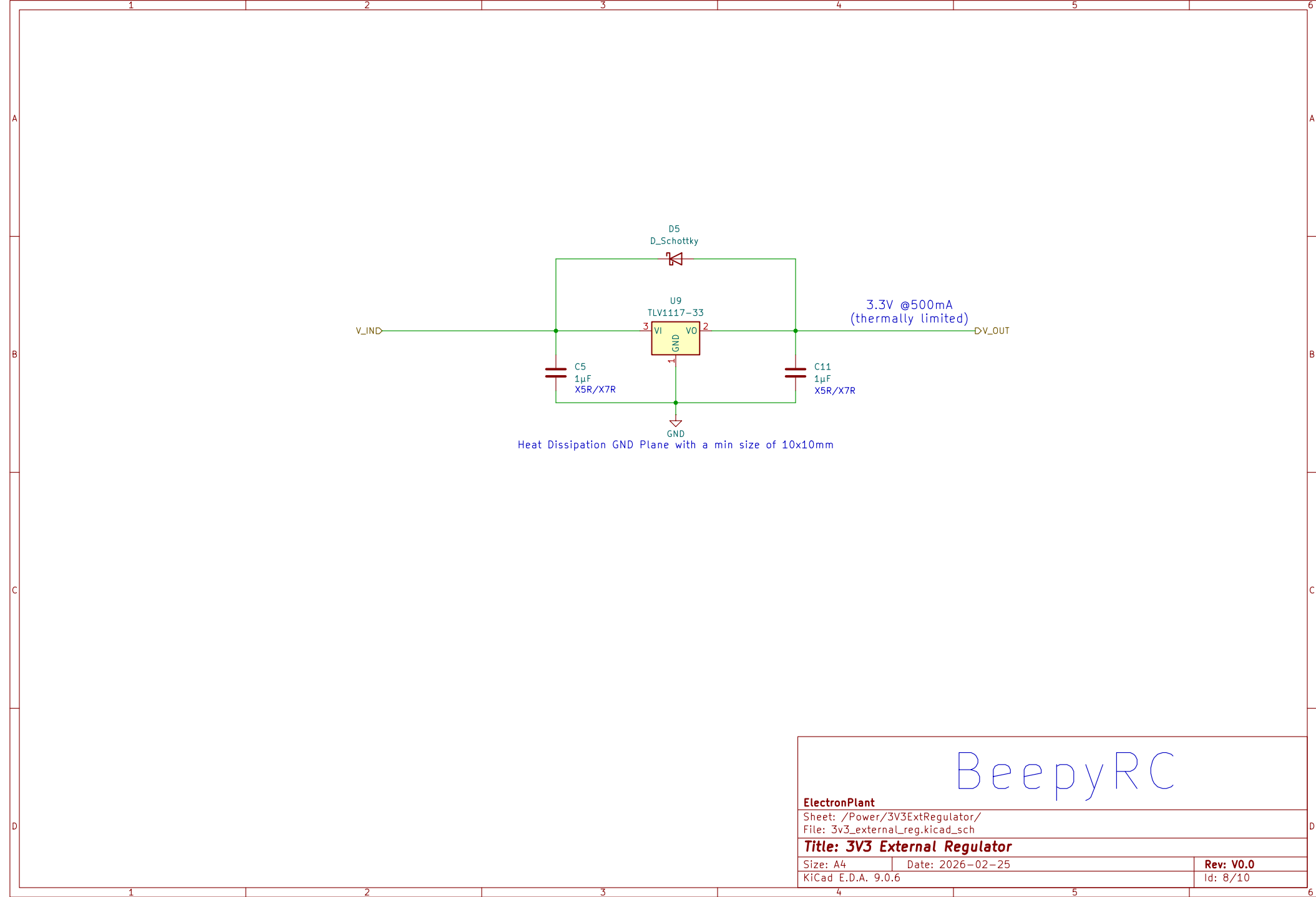
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Date: 2026-02-25

Rev: V0.0

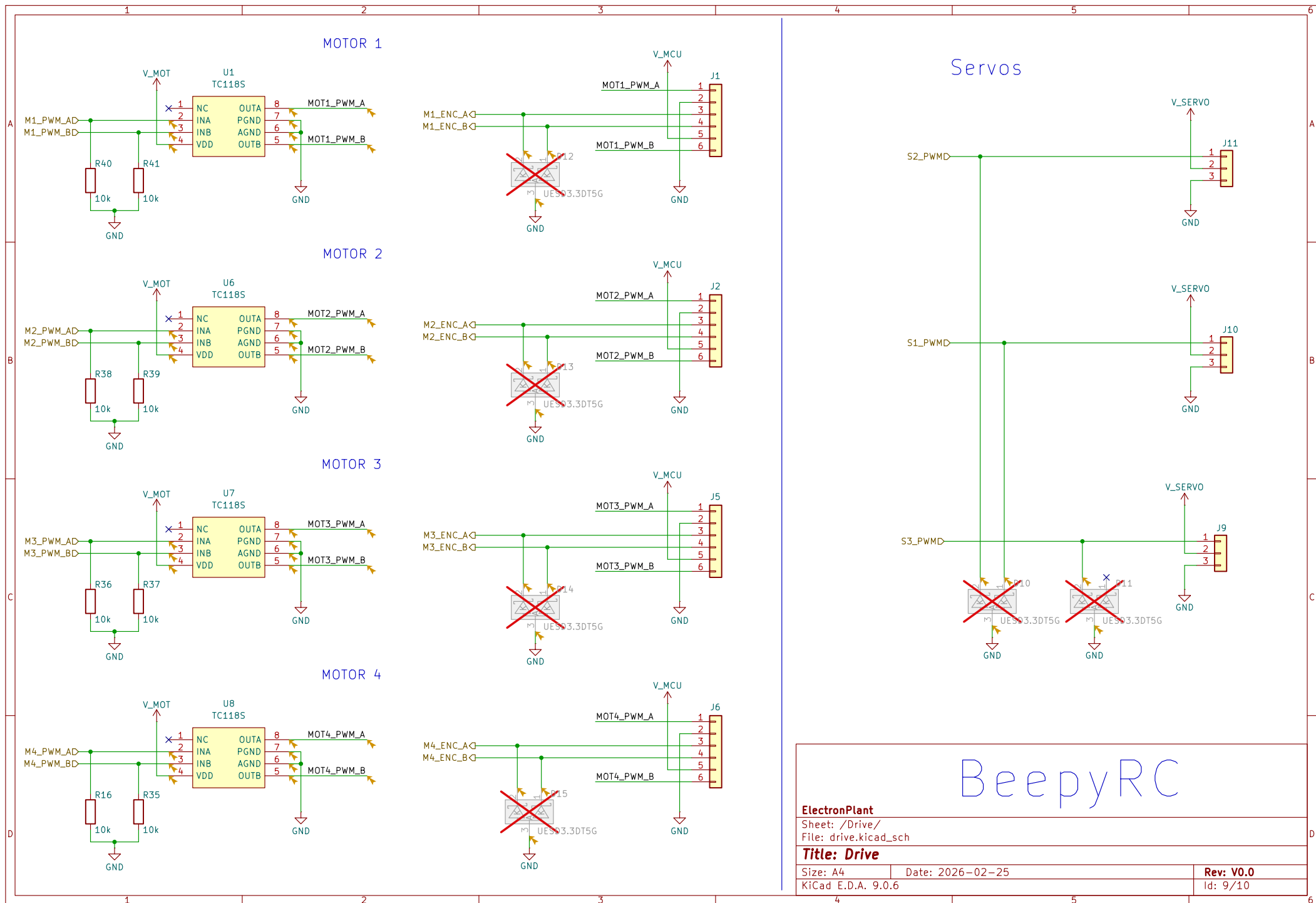
KiCad E.D.A. 9.0.6

Id: 8/10



BeepyRC

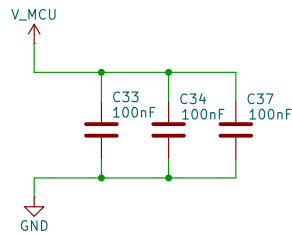
ElectronPlant		
Sheet: /Power/3V3ExtRegulator/		
File: 3v3_external_reg.kicad_sch		
Title: 3V3 External Regulator		
Size: A4	Date: 2026-02-25	Rev: V0.0
KiCad E.D.A. 9.0.6		Id: 8/10



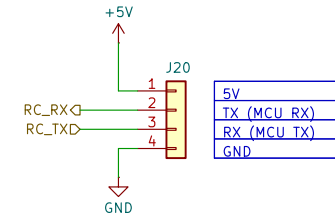
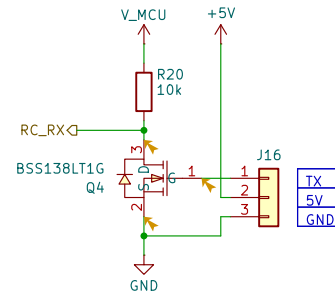
OSD

Decoupling strategy

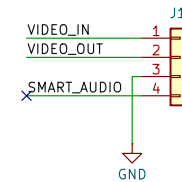
C14 between AVDD (P12) and AGND (P10)
C29 between DVDD (P2) and DGND (P1)
C30 between PVDD (P14) and PGND (P11)



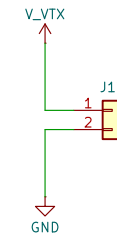
RC Serial



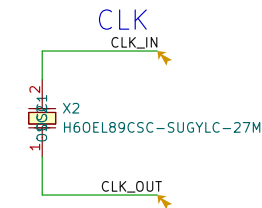
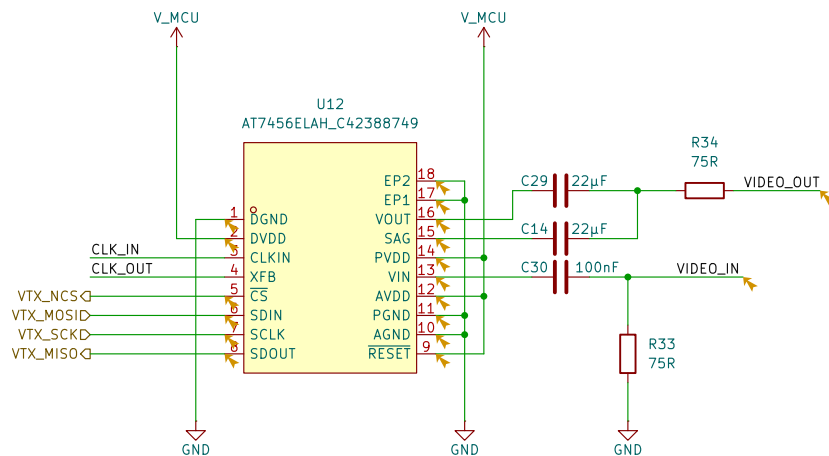
VTX



VTX Connector



Power Connector



BeepyRC

ElectronPlant

Sheet: /RC/
File: rc.kicad_sch

Title: RC

Size: A4
KiCad E.D.A. 9.0.6

Date: 2026-02-25

Rev: V0.0

Id: 10/10